

#1.write a python program to display "Hello, World!"

```
print("Hello,World!")
```

```
Hello,World!
```

#2.Write a python program to display your name and age

```
Name="Paramesh Perisetti"
```

```
Age=31
```

```
print("Name:",Name)
```

```
print("Age:",Age)
```

```
Name: Paramesh Perisetti
```

```
Age: 31
```

#3.print all keywords

```
Import keyword
```

```
print("Python keywords are....")
```

```
print(keyword.kwlist)
```

```
Python keywords are....  
['False', 'None', 'True', 'and', 'as', 'assert', 'async', 'await', 'break', 'class',  
'continue', 'def', 'del', 'elif', 'else', 'except', 'finally', 'for', 'from',  
'global', 'if', 'import', 'in', 'is', 'lambda', 'nonlocal', 'not', 'or', 'pass',  
'raise', 'return', 'try', 'while', 'with', 'yield']
```

#4.Check the given word is python keyword

```
Import keyword
```

```
x=input("enter any word")
```

```
if keyword.iskeyword(x):
```

```
    print("Your entered word is keyword in python")
```

```
else:
```

```
    print ("Your entered word is not a keyword in python")
```

```
enter any word break
```

```
Your entered word is keyword in python
```

#5.create list and tuple to change element differently for each

```
#create a list
```

```
my_list=[1,2,3]
```

```
print("Original list:",my_list)
```

```
#change element in a list
```

```
my_list[1]=4
```

```
print("changed list:",my_list)
```

```
#create a tuple
```

```
my_tuple=(1,2,3)
```

```
print("Original tuple:",my_tuple)
```

```
#change element in a tuple
```

```
my_tuple[2]=5
```

```
print("changed tuple:",my_tuple)
```

```
Original list: [1, 2, 3]  
changed list: [1, 4, 3]  
Original tuple: (1, 2, 3)
```

6.demonstrate mutable and immutable arguments

```
def demonstrate_mutability():
```

```
    # Immutable argument (integer)
```

```
    def modify_immutable(x):
```

```
        print(f"Original id of x: {id(x)}")
```

```
        x += 1
```

```
        print(f"Modified id of x inside function: {id(x)}")
```

```
    return x
```

```
num = 10

print(f"Before modify_immutable: {num}")

num = modify_immutable(num)

print(f"After modify_immutable: {num}")
```

```
# Mutable argument (list)

def modify_mutable(lst):

    print(f"Original id of lst: {id(lst)}")

    lst.append(4)

    print(f"Modified id of lst inside function: {id(lst)}")

    return lst
```

```
my_list = [1, 2, 3]

print(f"Before modify_mutable: {my_list}")

my_list = modify_mutable(my_list)

print(f"After modify_mutable: {my_list}")
```

```
demonstrate_mutability()

Before modify_immutable: 10

Original id of x: 140732544854744

Modified id of x inside function: 140732544854776

After modify_immutable: 11

Before modify_mutable: [1, 2, 3]

Original id of lst: 2832387070336

Modified id of lst inside function: 2832387070336
```

After modify_mutable: [1, 2, 3, 4]

#7.Basic arithmetic operations

```
def basic_arithmetic_operations():
```

```
    try:
```

```
        # Input: Taking two numbers from the user
```

```
        num1 = float(input("Enter the first number: "))
```

```
        num2 = float(input("Enter the second number: "))
```

```
        # Performing arithmetic operations
```

```
        addition = num1 + num2
```

```
        subtraction = num1 - num2
```

```
        multiplication = num1 * num2
```

```
        division = num1 / num2 if num2 != 0 else "undefined (division by zero)"
```

```
        floor_division = num1 // num2 if num2 != 0 else "undefined (division by zero)"
```

```
        modulus = num1 % num2 if num2 != 0 else "undefined (division by zero)"
```

```
        exponentiation = num1 ** num2
```

```
        # Displaying the results
```

```
        print(f"\nResults:")
```

```
        print(f"Addition: {num1} + {num2} = {addition}")
```

```
        print(f"Subtraction: {num1} - {num2} = {subtraction}")
```

```
        print(f"Multiplication: {num1} * {num2} = {multiplication}")
```

```
        print(f"Division: {num1} / {num2} = {division}")
```

```
        print(f"Floor Division: {num1} // {num2} = {floor_division}")
```

```
print(f"Modulus: {num1} % {num2} = {modulus}")

print(f"Exponentiation: {num1} ** {num2} = {exponentiation}")
```

```
except ValueError:
```

```
    print("Invalid input! Please enter numeric values.")
```

```
# Call the function to execute
```

```
basic_arithmetic_operations()
```

```
Enter the first number: 2
Enter the second number: 3

Results:
Addition: 2.0 + 3.0 = 5.0
Subtraction: 2.0 - 3.0 = -1.0
Multiplication: 2.0 * 3.0 = 6.0
Division: 2.0 / 3.0 = 0.6666666666666666
Floor Division: 2.0 // 3.0 = 0.0
Modulus: 2.0 % 3.0 = 2.0
Exponentiation: 2.0 ** 3.0 = 8.0
```

#8.logical operators

```
def demonstrate_logical_operators():
```

```
    # Define sample variables
```

```
    age = 25
```

```
    is_student = True
```

```
    has_discount_coupon = False
```

```
    is_sunny = True
```

```
    is_weekend = False
```

```
    # AND operator: True if both conditions are true
```

```
    if age > 18 and is_student:
```

```
print("Eligible for student discount.")
```

```
# OR operator: True if at least one condition is true
```

```
if has_discount_coupon or is_student:
```

```
    print("Eligible for discount.")
```

```
# NOT operator: Reverses the boolean value
```

```
if not is_weekend:
```

```
    print("It's a weekday.")
```

```
# Combining logical operators
```

```
if (is_sunny and is_weekend) or (is_student and not is_weekend):
```

```
    print("Ideal conditions for an outdoor activity.")
```

```
# Call the function to execute
```

```
demonstrate_logical_operators()
```

```
Eligible for student discount.  
Eligible for discount.  
It's a weekday.  
Ideal conditions for an outdoor activity.
```

#9.string to integer,float,boolean

```
# Converting a string into an integer
```

```
number_str = "10"
```

```
number_int = int(number_str)
```

```
print(number_int)
```

```
# Converting a string into a float
```

```
decimal_str = "10.5"
```

```
decimal_float = float(decimal_str)
```

```
print(decimal_float)
```

```
s = "True"
```

```
# Convert string to boolean using eval
```

```
boolean_value = eval(s)
```

```
print(boolean_value)
```

```
10  
10.5  
True
```

#10.

```
# Mixed-type list: strings, floats, and integers
```

```
mixed_list = ['10', '20.5', 30, 40.7, '50']
```

```
print("Original list:", mixed_list)
```

```
# Convert all elements to float
```

```
float_list = [float(x) for x in mixed_list]
```

```
print("Converted to float:", float_list)
```

```
# Convert all elements to integer (note: this will truncate floats)
```

```
int_list = [int(float(x)) for x in mixed_list]
```

```
print("Converted to int:", int_list)
```

```
# Convert all elements to string
```

```
str_list = [str(x) for x in mixed_list]

print("Converted to string:", str_list)
```

```
Original list: ['10', '20.5', 30, 40.7, '50']
Converted to float: [10.0, 20.5, 30.0, 40.7, 50.0]
Converted to int: [10, 20, 30, 40, 50]
Converted to string: ['10', '20.5', '30', '40.7', '50']
```

11.Input from the user

```
num = float(input("Enter a number: "))
```

```
# Check conditions
```

```
if num > 0:
```

```
    print("The number is positive.")
```

```
elif num < 0:
```

```
    print("The number is negative.")
```

```
else:
```

```
    print("The number is zero.")
```

#12.

```
for i in range(1, 11):
```

```
    print(i)
```

```
1
2
3
4
5
6
7
8
9
10
```

#13.


```
# Initialize sum

even_sum = 0


# Loop through numbers from 1 to 50
for num in range(1, 51):

    if num % 2 == 0:

        even_sum += num


# Print the result
print("The sum of even numbers from 1 to 50 is:", even_sum)
```

```
The sum of even numbers from 1 to 50 is: 650
```

#14.

```
# Input string
original = input("Enter a string: ")


# Initialize variables
reversed_str = ""
index = len(original) - 1


# Loop backwards through the string
while index >= 0:

    reversed_str += original[index]

    index -= 1
```

```
# Print the reversed string
```

```
print("Reversed string:", reversed_str)
```

```
Enter a string: paramesh  
Reversed string: hsemarap
```

#15.

```
# Get input from the user
```

```
num = int(input("Enter a non-negative integer: "))
```

```
# Check for negative input
```

```
if num < 0:
```

```
    print("Factorial is not defined for negative numbers.")
```

```
else:
```

```
    factorial = 1
```

```
    n = num
```

```
    while n > 1:
```

```
        factorial *= n
```

```
        n -= 1
```

```
# Print the result
```

```
print(f"The factorial of {num} is {factorial}.")
```

```
Enter a non-negative integer: 5  
The factorial of 5 is 120.
```